

BPE Tokenization of the Same Sentence Across Languages (LLaMA-3 Tokenizer)

(Fertility F = number of subword tokens produced per sentence)

English (Latin)

"The scientist discovered a new species"

The	scientist	discovered	a	new	species
-----	-----------	------------	---	-----	---------

F = 6

German (Latin)

"Der Wissenschaftler entdeckte eine neue Art"

Der	Wissen	schaft	ler	ent	deck	te	eine	neue	Art
-----	--------	--------	-----	-----	------	----	------	------	-----

F = 10

Amharic (Ge'ez)

"ሳይንቲስቱ አዲስ ዝርያ አገኘ"

(LLaMA-3 Byte-Level BPE Fallback)

<0xE1>	<0x88>	<0xB3>	<0xE1>	<0x8B>	<0xAD>	<0xE1>	<0x8A>	<0x95>	<0xE1>	<0x89>	<0xB2>	<0xE1>	...
--------	--------	--------	--------	--------	--------	--------	--------	--------	--------	--------	--------	--------	-----

F = 48

Higher fertility
(Fragmentation)

* Note: LLaMA-3's BPE vocabulary contains no merges for Ge'ez script; each 3-byte character decomposes into 3 raw UTF-8 byte tokens (<0x..>), yielding F = 48 tokens.